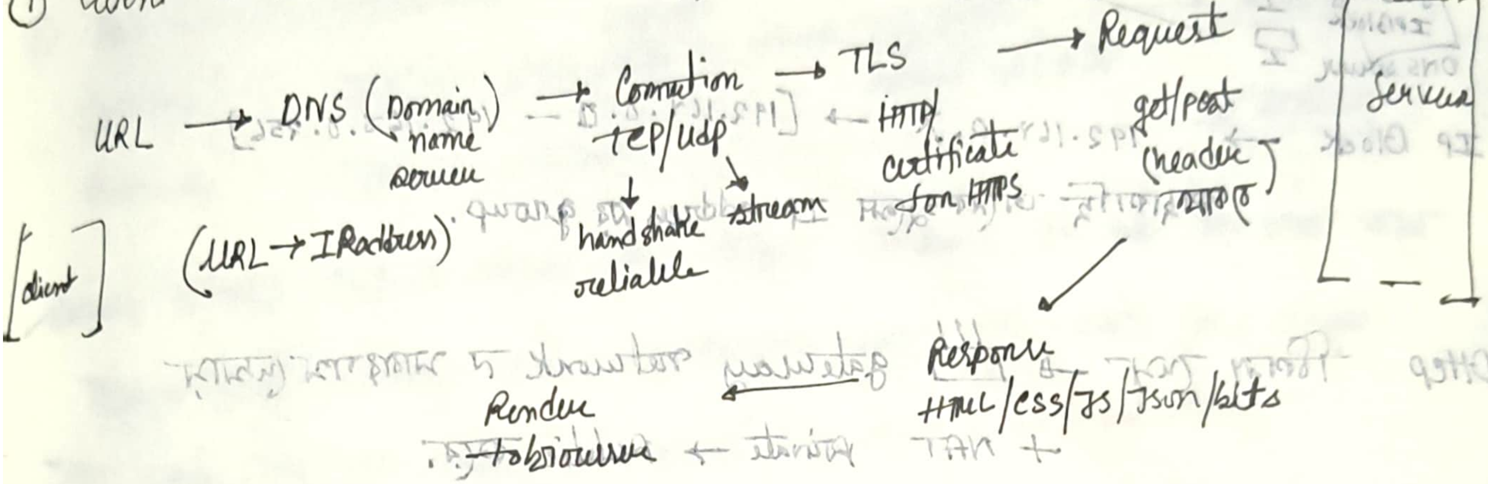
Day 01 : networking, DNS, HTTP/S, Routing

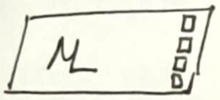
① client to server basic Req/Res.


URL
DNS (Domain name server)
(URL → IPaddress)
client
Connection TCP/UDP
handshake reliable
stream
TLS
HTTP certificate for HTTPS
Request
get/post (header আলাদা)
Server
Response HTML/CSS/JS/Json/bits
Render
~~to browser~~

1.1) IP

→ IPV4 [32 bit] [4.3 billion] [192.168.1.10]

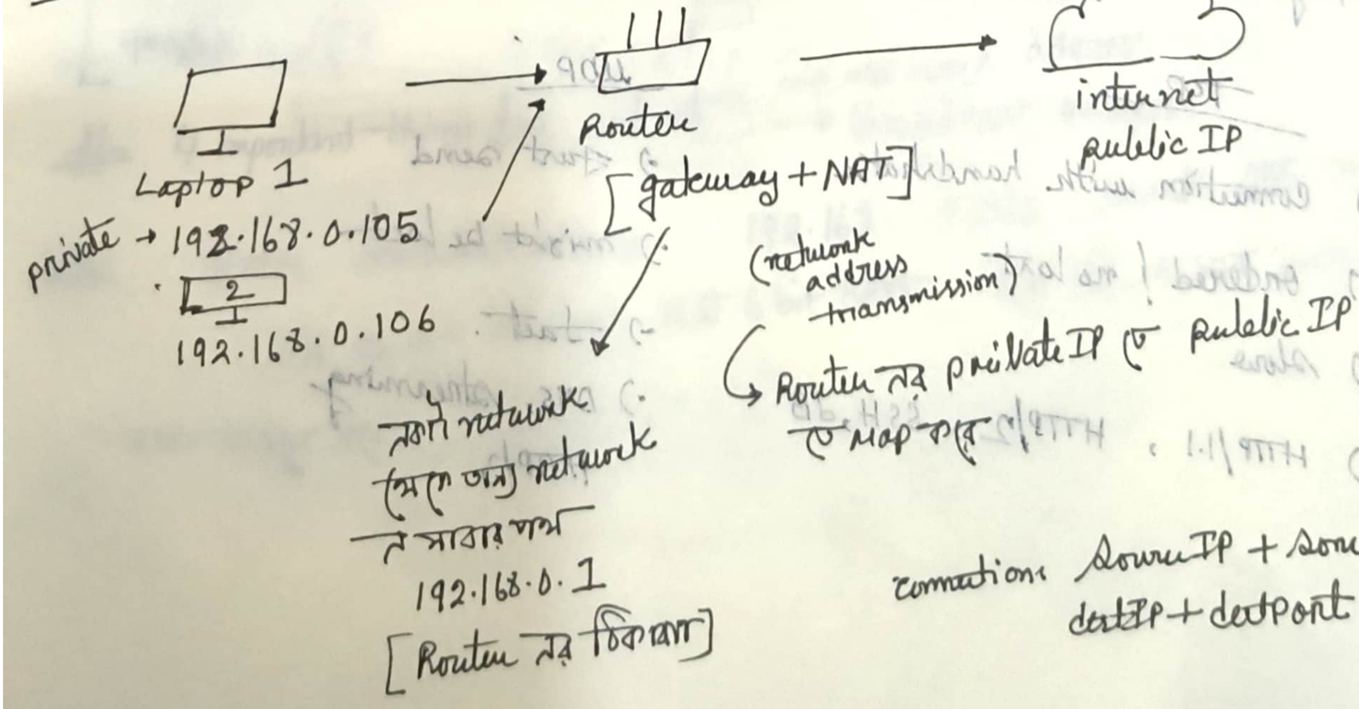
→ IPV6 [128 bit] [Large] [2001:db8::10]



each machine has unique IP ~~port numbers~~ and port for hit

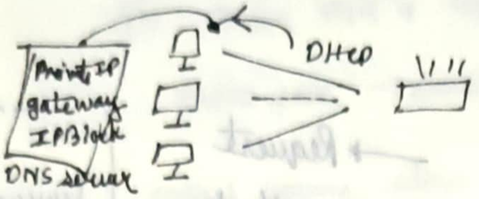
HTTP 80
SSH 22
HTTPS 443

1.2) private and public IP


Laptop 1
private → 192.168.0.105
2
192.168.0.106
Router
[gateway + NAT]
(network address transmission)
internet
public IP
Router এর private IP কে public IP তে map করে
বাকী network (মানে অন্য network) এ যাবার পথ
192.168.0.1
[Router এর ঠিকানা]

connection: source IP + source port + destIP + destPort + protocol

- DHCP network এ যুক্ত হওয়া device কে Automatic কিছু info দেয়



- IP Block → 192.168.0.X → [192.168.0.0 — 192.168.0.256]

  কাছাকাছি অনেকগুলো IP address এর group.

DHCP ঠিকানা দেয় → gateway network এ যোগাযোগ দেখায়

+ NAT private → public করে.

2) Network layer

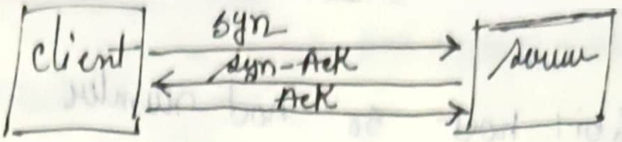
OSI

Application
Presentation
Session — gmail, SMTP, HTTP/HTTPS, DNS
Transport — TCP, UDP.
Network
Datalink — IP, MAC Address
physical — wire, ethernet, wifi.

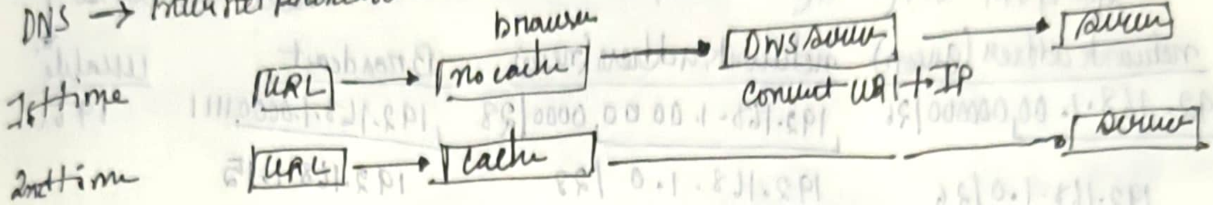
2.1)

TCP	UDP
·) Connection with handshake	·) Just send
·) Ordered / no lost	·) might be lost
·) slow	·) fast.
·) HTTP/1.1, HTTP/2, SSH, dB	·) DNS, streaming HTTP/3

handshake in TCP :


client
syn
syn-Ack
Ack
server

3) DNS → Internet phonebook


1st time
URL
browser
no cache
DNS server
Convert URL to IP
server
2nd time
URL
cache
server

server change করলে IP change হবে, কিন্তু browser IP cache করে রাখে মানে পুরোনো IP তে hit করবে। তাই Migration এর আগে TTL কমিয়ে রাখা হলো যে cache হবে কম সময়ের জন্য, বারবার DNS server কে check করবে।

TTL কমিয়ে রাখবো কেন? ↑

③ ▫ network subnetting

IPV4 → 32 bit → 8 bit (0-255) 8 (0-255) 8 (0-255) 8 (0-255)

এর prefix হলো কতটুকু network আর বাকিটা host.

যেমন IP : 192.168.1.70
subnetmask : 255.255.255.0
prefix : /24

→ এখানে 24 bit network বাকি 8 bit host

4 Important things for subnet → network address → Broadcast address.

Ex: given network address 192.168.1.0/26, mask 255.255.255.192

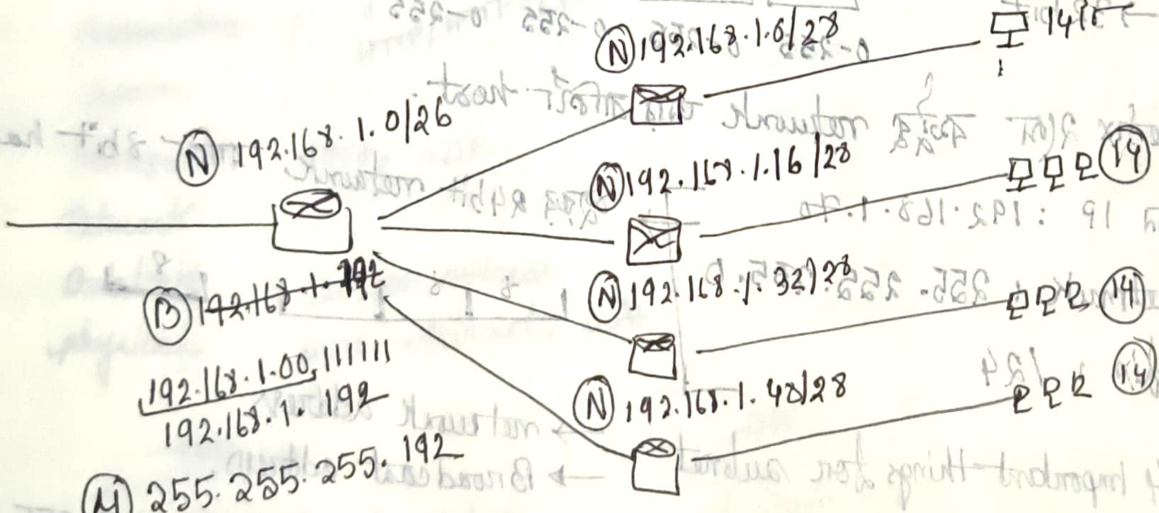
এর 26 bit network last 6 bit host থেকে 4টা সমান subnet এ

ভাগ করতে হবে।

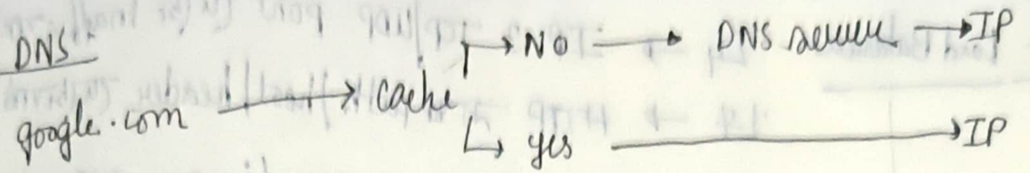
⇒ last 6bit host তো host number $2^6 = 64$

তাকে 4ভাগ করতে হবে $64/4 = 16$। যার প্রথম ও শেষ network ও host address so] ↓

subnet	network address (given)	network address (new)	Broadcast	usable
①	192.168.1.00,000000/26 192.168.1.0/26	192.168.1.0000,0000/28 192.168.1.0/28	192.168.1.0000,1111 192.168.1.15	14টি
②		192.168.1.0001,0000/28 192.168.1.16/28	192.168.1.0001,1111 192.168.1.31	14টি
③		192.168.1.0010,0000/28 192.168.1.32/28	192.168.1.0010,1111 192.168.1.47	14টি 14
④		192.168.1.0011,0000/28 192.168.1.48/28	192.168.1.0011,1111 192.168.1.63	14


(N) 192.168.1.0/26
(B) ~~192.168.1.192~~
192.168.1.00,111111
192.168.1.192
(M) 255.255.255.192
(N) 192.168.1.0/28
14টি
(N) 192.168.1.16/28
(14)
(N) 192.168.1.32/28
(14)
(N) 192.168.1.48/28
(14)

~~DNS~~ DNS


google.com
cache
NO
DNS server
IP
yes
IP

TTL বলে DNS IP browser কতক্ষন মনে রাখবে,

DNS Record type

A → IPV4
AAAA → IPV6
CNAME → alias (anothername) www.google.com google.com
MX → mail server @google.com
TXT → Free text
NS → domains name server

④ HTTP

Req

Get /about HTTP/1.1 [Methode+path/statuscode]
Host: google.com [headers]
Accept: text/html/json
Cookie: ~~ø~~
<body>

Res

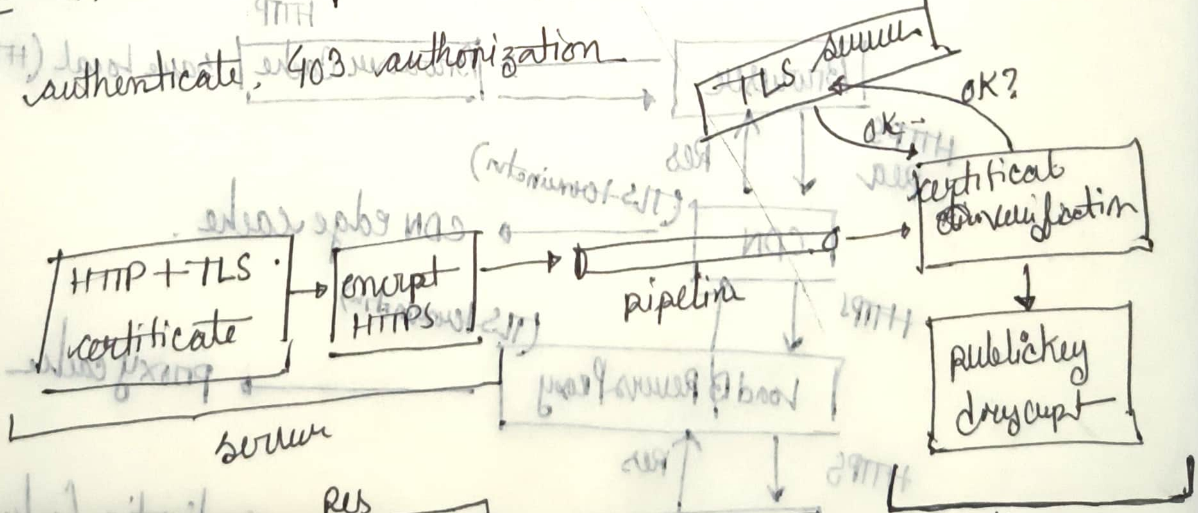
HTTP/1.1 200 OK
contenttype : text/html/json
set cookie :
<body>

Status Code

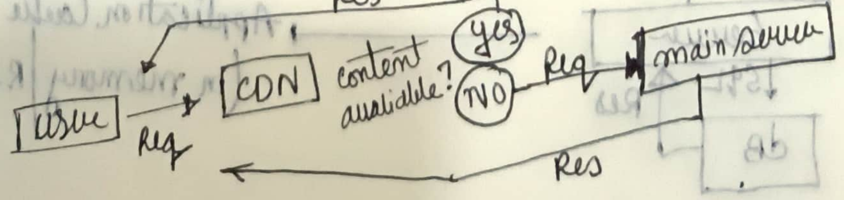
2XX success | 3XX go elsewhere | 4XX clientMistake | 5XX server problem

401 authenticate, 403 authorization

HTTPS


HTTP + TLS certificate
encrypt HTTPS
pipeline
server
TLS server
OK?
OK
certificat conVerification
publickey decrypt
client

CDN

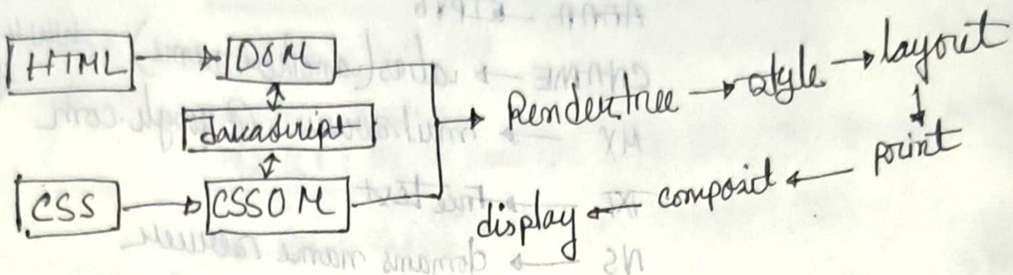

user
Req
CDN
content available?
yes
NO
Req
main server
Res
Res

L4 VS L7 Load Balancer L4 → IP ও TCP/UDP port দেখে traffic পাঠায়

L7 → HTTP এর path/host/header দেখে পাঠায়

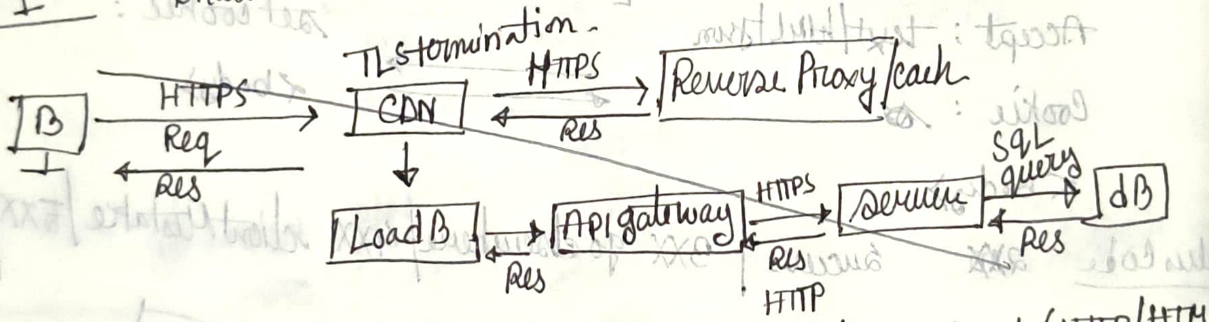
যেমন /api গেলে server এ /image অন্য server এ

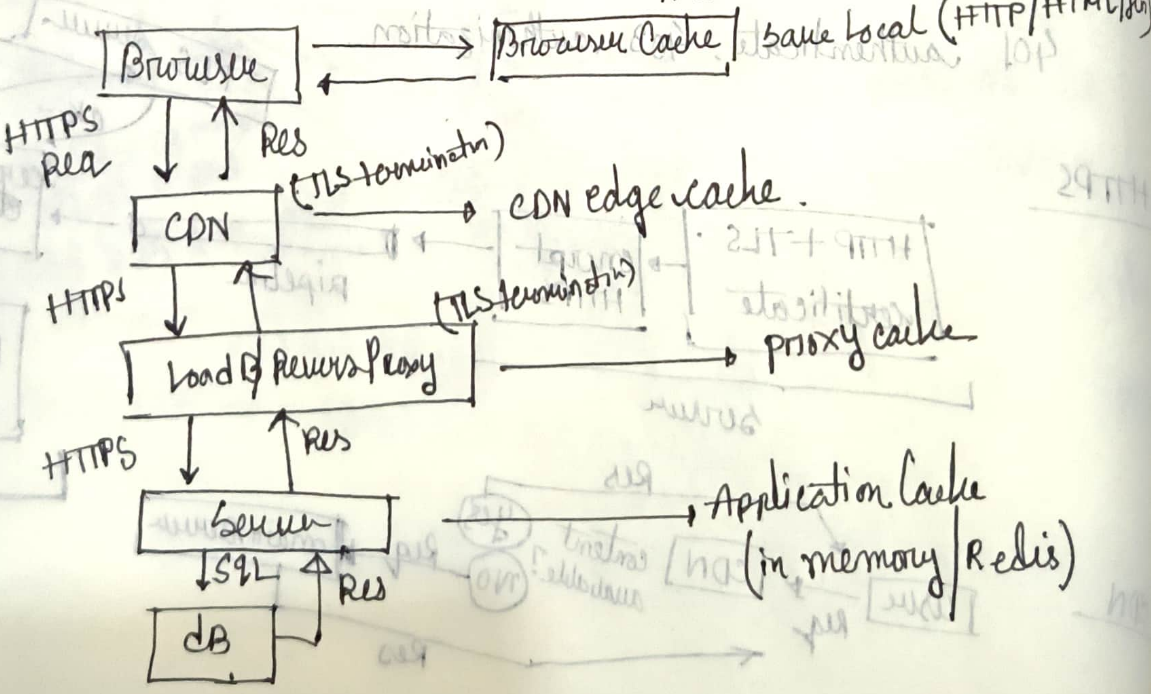
⑤ Server Response HTML/css browser have to draw it


HTML
DOM
Javascript
CSS
CSSOM
Render tree
style
layout
paint
composit
display

CSS should load 1st then html then JS.

Task 1 : Draw the full Request path from browser to dB


TLS termination.
B
HTTPS
Req
Res
CDN
HTTPS
Res
Reverse Proxy/cache
Load B
Res
Api gateway
HTTPS
Res
HTTP
server
SQL query
Res
dB


Browser
Browser Cache
save Local (HTTP/HTML/Json)
HTTPS Req
Res
CDN
(TLS termination)
CDN edge cache.
HTTPS
Load B Reverse Proxy
(TLS termination)
Proxy cache
HTTPS
Res
Server
Application Cache (in memory/Redis)
SQL
Res
dB